

CARLOS J FELIX VELEZ

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PROFESSIONAL EXPERIENCE

PENTAIR – Cary, NC

AUTOMATION & ROBOTICS ENGINEER | *Advanced Manufacturing & Technology Group* | Mar 2023 – Present

- Lead automation strategy, system design, and project execution across six manufacturing sites. Created standardized deployment frameworks to guide project teams from Ideation to Site Acceptance Testing.
- Acted as a cross-functional leader between site operations, enterprise teams, and external integrators – shaping 5-year automation roadmaps and driving standardization, budget alignment, and execution.
- Acted as robotics and automation SME and managed manufacturing engineers across 6 sites, guiding them to deliver capital projects in automated assembly, inspection, packaging and autonomous mobile robots (AMRs).

CORNING, INC – Durham, NC

AUTOMATION ENGINEER | *Manufacturing, Technology & Engineering* | Nov 2021 – Mar 2023

- Led full CQV project lifecycle for an automated pipet assembly complex, applying Six Sigma and validation protocols to reduce downtime by 50% and generate projected savings of over \$1M per year.
- Programmed FANUC robot error-recovery methods for a glass optical fiber production cell and used a 2-D vision system for robot path planning to improve system robustness, reduce downtime, and enhance operator safety.
- Designed and developed cost-effective automation prototypes, collaborating with site operations stakeholders to improve production processes and overall equipment performance.

BOEING – St Louis, MO

MECHANICAL SYSTEMS DESIGN & ANALYSIS ENGINEER | *Boeing Global Services* | Jul 2020 – Nov 2021

- Developed dynamics simulations in MATLAB and Simcenter 3D to model battery performance and steering behavior for a robotic vehicle designed to lift and tow a fighter jet on a military aircraft carrier.
- Solved legacy equipment obsolescence for ground support equipment by identifying and validating modern alternatives for sensors, motors, and oscillators under stringent performance specs.

MANUFACTURING SYSTEMS & SIMULATION ENGINEER | *Boeing Research & Technology* | Aug 2018 – Jul 2020

- Coordinated a \$925k robotic composite lay-up research project as Principal Investigator with two university partners. Developed the lay-up tool surface, designed quality inspection requirements and oversaw composite lay-up experiments using ABB robots.
- Developed and tested prototypes of automated solutions for manufacturing processes including motorized gantries, wireless smart tools, and collaborative robot work cells in a lab environment.

EDUCATION

RENSSELAER POLYTECHNIC INSTITUTE – Troy, NY

Master of Engineering in Mechanical Engineering with focus in Control Systems | May 2018

Bachelor of Architecture | May 2017

Purdue University Online: Lean Six Sigma Green Belt Certification | May 2021

SKILLS

Programming: Python, C++, PLC programming (Rockwell), MATLAB

CAD: Solidworks, AutoCAD, Siemens NX, Rhino 3D

Robotics: FANUC robot programming, Robot Operating System (ROS), ANSI/RIA Standards

Project Management: Scaled Agile Framework (SAFe), Equipment Validation (IQ/OQ/PQ), DFMA

Technical: Functional design specifications, hardware-in-the-loop testing, vision inspection (Keyence)